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## 1) Deriving the equations of motion

### 1.1) Definitions

$l_1$  = length of main pendulum rod

$l_2$  = length of second pendulum rod

$\theta$  = angle made by main pendulum rod with the vertical

$\phi$  = angle made by the second pendulum rod with the vertical

$m_1$  = mass of main pendulum

$m_2$  = mass of second pendulum

### 1.2) Kinematics

Position vector of the main pendulum:

$$\mathbf{s}_1(\theta) = l_1(\sin \theta, \cos \theta)$$

Velocity of the main pendulum:

$$\begin{aligned}\mathbf{v}_1 &= l_1 \dot{\theta}(\cos \theta, -\sin \theta) \\ |\mathbf{v}_1|^2 &= l_1^2 \dot{\theta}^2\end{aligned}$$

Position vector of the second pendulum:

$$\mathbf{s}_2(\theta, \phi) = (l_1 \sin \theta + l_2 \sin \phi, l_1 \cos \theta + l_2 \cos \phi)$$

Velocity vector of the second pendulum:

$$\begin{aligned}\mathbf{v}_2 &= (l_1 \dot{\theta} \cos \theta + l_2 \dot{\phi} \cos \phi, -l_1 \dot{\theta} \sin \theta - l_2 \dot{\phi} \sin \phi) \\ |\mathbf{v}_2|^2 &= l_1^2 \dot{\theta}^2 + l_2^2 \dot{\phi}^2 + 2l_1 l_2 \dot{\theta} \dot{\phi} \cos(\theta - \phi)\end{aligned}$$

### 1.3) Dynamics

Kinetic energy of the system is:

$$T = \frac{m_1}{2} l_1^2 \dot{\theta}^2 + \frac{m_2}{2} \left[ l_1^2 \dot{\theta}^2 + l_2^2 \dot{\phi}^2 + 2l_1 l_2 \dot{\theta} \dot{\phi} \cos(\theta - \phi) \right]$$

Potential energy of the system is:

$$U = -m_1 g l_1 \cos \theta - m_2 g (l_1 \cos \theta + l_2 \cos \phi)$$

Lagrangian of the system is:

$$\begin{aligned}L &= T - U \\ L &= \frac{m_1}{2} l_1^2 \dot{\theta}^2 + \frac{m_2}{2} \left[ l_1^2 \dot{\theta}^2 + l_2^2 \dot{\phi}^2 + 2l_1 l_2 \dot{\theta} \dot{\phi} \cos(\theta - \phi) \right] + m_1 g l_1 \cos \theta + m_2 g (l_1 \cos \theta + l_2 \cos \phi)\end{aligned}$$

### 1.4) Euler-Lagrange equations

Equation for  $\theta$ :

$$\begin{aligned}\frac{\partial L}{\partial \theta} &= -m_2 l_1 l_2 \dot{\theta} \dot{\phi} \sin(\theta - \phi) - g l_1 (m_1 + m_2) \sin \theta \\ \frac{\partial L}{\partial \dot{\theta}} &= (m_1 + m_2) l_1^2 \dot{\theta} + m_2 l_1 l_2 \dot{\phi} \cos(\theta - \phi)\end{aligned}$$

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{\theta}} \right) = (m_1 + m_2) l_1^2 \ddot{\theta} + m_2 l_1 l_2 \ddot{\phi} \cos(\theta - \phi) - m_2 l_1 l_2 \dot{\theta} \dot{\phi} \sin(\theta - \phi) + m_2 l_1 l_2 \dot{\phi}^2 \sin(\theta - \phi)$$

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{\theta}} \right) = \frac{\partial L}{\partial \theta}$$

$$\ddot{\theta} + \frac{m_2}{m_1 + m_2} \frac{l_2}{l_1} \left( \ddot{\phi} \cos(\theta - \phi) + \dot{\phi}^2 \sin(\theta - \phi) \right) + \frac{g}{l_1} \sin \theta = 0 \quad (1)$$

Equation for  $\phi$ :

$$\begin{aligned}\frac{\partial L}{\partial \phi} &= m_2 l_1 l_2 \dot{\theta} \dot{\phi} \sin(\theta - \phi) - m_2 g l_2 \sin \phi \\ \frac{\partial L}{\partial \dot{\phi}} &= m_2 l_2^2 \dot{\phi} + m_2 l_1 l_2 \dot{\theta} \cos(\theta - \phi)\end{aligned}$$

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{\phi}} \right) = m_2 l_2^2 \ddot{\phi} + m_2 l_1 l_2 \ddot{\theta} \cos(\theta - \phi) - m_2 l_1 l_2 \dot{\theta}^2 \sin(\theta - \phi) + m_2 l_1 l_2 \dot{\theta} \dot{\phi} \sin(\theta - \phi)$$

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{\phi}} \right) = \frac{\partial L}{\partial \phi}$$

$$\ddot{\phi} + \frac{l_1}{l_2} \left( \ddot{\theta} \cos(\theta - \phi) - \dot{\theta}^2 \sin(\theta - \phi) \right) + \frac{g}{l_2} \sin \phi = 0 \quad (2)$$

## 1.5) Converting the system of second order ODEs to a system of first order ODEs

Equations (1) and (2) can be represented in matrix form as:

$$\begin{pmatrix} 1 & \mu \frac{l_2}{l_1} \cos(\Delta) \\ \frac{l_1}{l_2} \cos(\Delta) & 1 \end{pmatrix} \begin{pmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \end{pmatrix} = \begin{pmatrix} -\mu \frac{l_2}{l_1} \omega_2^2 \sin(\Delta) - \frac{g}{l_1} \sin \theta \\ \frac{l_1}{l_2} \omega_1^2 \sin(\Delta) - \frac{g}{l_2} \sin \phi \end{pmatrix}$$

Where  $\frac{m_2}{m_1+m_2} = \mu$ ,  $\theta - \phi = \Delta$ ,  $\dot{\theta} = \omega_1$ ,  $\dot{\phi} = \omega_2$

Which upon solving gives:

$$\begin{pmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \end{pmatrix} = \begin{pmatrix} \frac{-\mu \frac{l_2}{l_1} \omega_2^2 \sin(\Delta) - \frac{g}{l_1} \sin \theta - \mu \omega_1^2 \cos(\Delta) \sin(\Delta) + \mu \frac{g}{l_1} \cos(\Delta) \sin \phi}{1 - \mu \cos^2(\Delta)} \\ \frac{\mu \omega_2^2 \cos(\Delta) \sin(\Delta) + \frac{g}{l_2} \cos(\Delta) \sin \theta + \frac{l_1}{l_2} \omega_1^2 \sin(\Delta) - \frac{g}{l_2} \sin \phi}{1 - \mu \cos^2(\Delta)} \end{pmatrix}$$

Let  $\mathbf{r} = (\theta, \omega_1, \phi, \omega_2)$  be the state vector of the system, then,

$$\frac{d\mathbf{r}}{dt} = (\dot{\theta}, \dot{\omega}_1, \dot{\phi}, \dot{\omega}_2)$$

$$\frac{d\mathbf{r}}{dt} = (\omega_1, \dot{\omega}_1(\mathbf{r}(t)), \omega_2, \dot{\omega}_2(\mathbf{r}(t)))$$

$$\frac{d\mathbf{r}}{dt} = \mathbf{F}(\mathbf{r}(t))$$

where

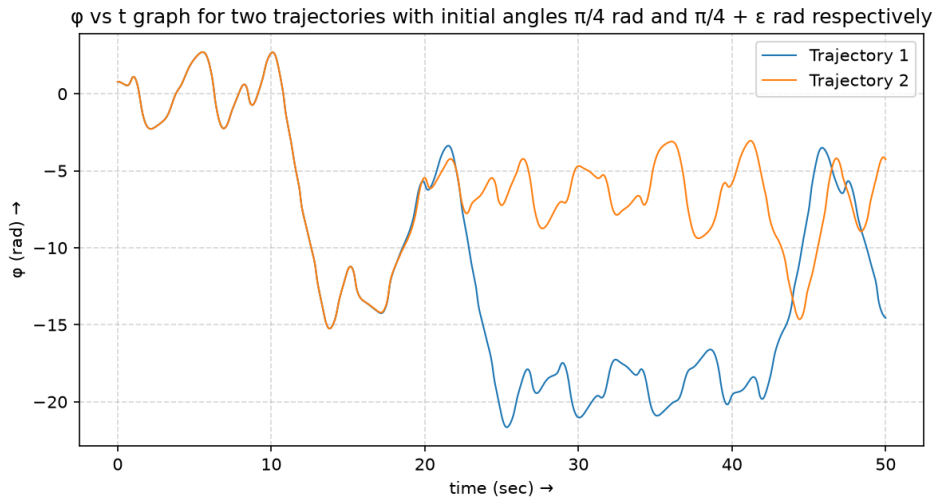
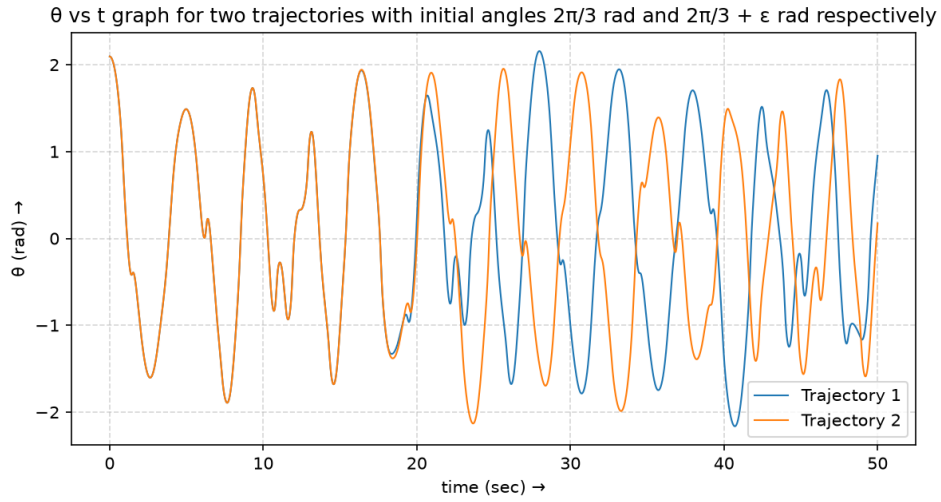
$$\mathbf{F}(\mathbf{r}(t)) = (\omega_1, \dot{\omega}_1(\mathbf{r}(t)), \omega_2, \dot{\omega}_2(\mathbf{r}(t)))$$

So we have successfully converted our system of two second order ODEs into four first order ODEs that can be numerically integrated using RK4.

## 2) Exploring what makes the system “chaotic”

### 2.1) Time evolution of two trajectories

Numerically integrating the system of ODEs for the initial state  $\mathbf{r}(0) = (\frac{2\pi}{3}, 0, \frac{\pi}{4}, 0)$  with  $(m_1, l_1, m_2, l_2) = (5, 3, 7, 2)$  and an infinitesimal perturbation of the initial state  $\mathbf{r}'(0) = \mathbf{r}(0) + \mathbf{h}(0)$ , where  $\mathbf{h}(0) = 10^{-6}(1, 0, 1, 0)$  allows us to plot angle vs time plots for the two trajectories. The plots for both  $\theta$  vs  $t$  and  $\phi$  vs  $t$  shows significant deviation between the two trajectories. This indicates sensitivity to initial conditions.



## 2.2) Characterizing the evolution of the perturbation vector

Let  $\mathbf{r}'(t) = \mathbf{r}(t) + \mathbf{h}(t)$  be another trajectory satisfying the ODE  $\frac{d\mathbf{r}'(t)}{dt} = \mathbf{F}(\mathbf{r}'(t))$ .

So,

$$\frac{d(\mathbf{r} + \mathbf{h})}{dt} = \mathbf{F}(\mathbf{r}(t) + \mathbf{h}(t))$$

$$\frac{d\mathbf{r}}{dt} + \frac{d\mathbf{h}}{dt} = \mathbf{F}(\mathbf{r}(t) + \mathbf{h}(t))$$

By performing a first order taylor expansion of  $\mathbf{F}(\mathbf{r}(t) + \mathbf{h}(t))$  about  $\mathbf{r}(t)$  we get:

$$\frac{d\mathbf{r}}{dt} + \frac{d\mathbf{h}}{dt} = \mathbf{F}(\mathbf{r}(t)) + [D\mathbf{F}(\mathbf{r}(t))]\mathbf{h}(t) + O(|\mathbf{h}|^2)$$

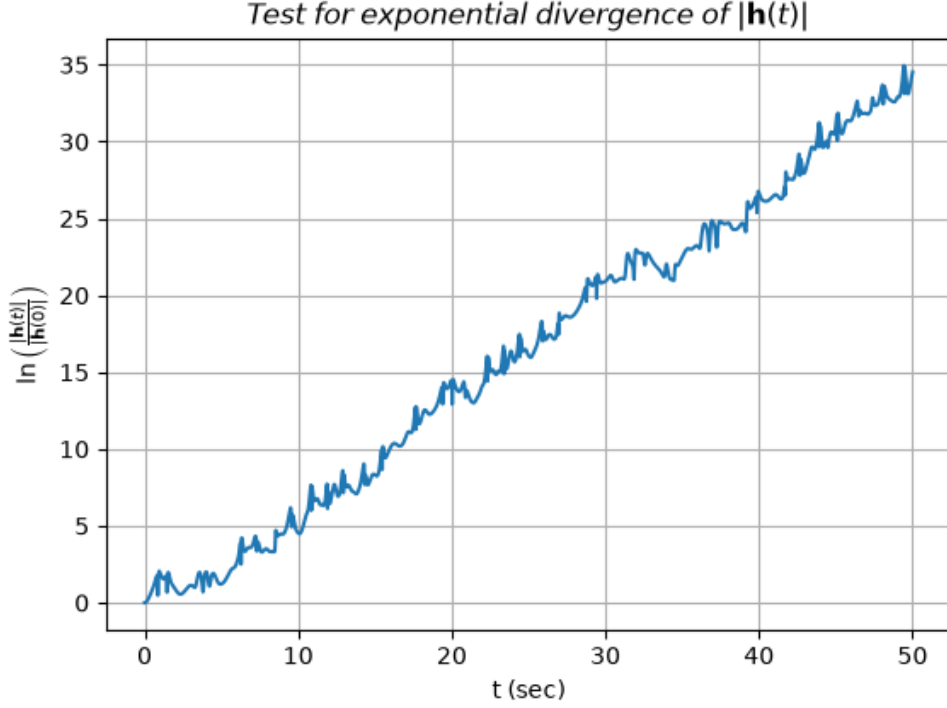
$$\frac{d\mathbf{h}}{dt} = [D\mathbf{F}(\mathbf{r}(t))]\mathbf{h}(t) + O(|\mathbf{h}|^2)$$

So upto first order our perturbation vector follows the ODE:

$$\frac{d\mathbf{h}}{dt} = [D\mathbf{F}(\mathbf{r}(t))]\mathbf{h}(t)$$

where  $[D\mathbf{F}(\mathbf{r}(t))]$  is the jacobian matrix of  $\mathbf{F}(\mathbf{r}(t))$ .

Numerically integrating both the perturbation vector ODE and the DP ODEs simultaneously for the initial state  $\mathbf{r}(0) = (\frac{2\pi}{3}, 0, \frac{\pi}{4}, 0)$  with  $\mathbf{h}(0) = 10^{-6}(1, 0, 1, 0)$  and  $(m_1, l_1, m_2, l_2) = (5, 3, 7, 2)$  allows us to plot the logarithmic plot of  $\ln\left(\frac{|\mathbf{h}(t)|}{|\mathbf{h}(0)|}\right)$  vs  $t$ . The obtained plot is quite jagged but is approximately linear.



The logarithmic plot is testing for exponential divergence of nearby trajectories in state space, for if:

$$|\mathbf{h}(t)| \approx |\mathbf{h}(0)|e^{\lambda t}$$

Then

$$\ln\left(\frac{|\mathbf{h}(t)|}{|\mathbf{h}(0)|}\right) \approx \lambda t$$

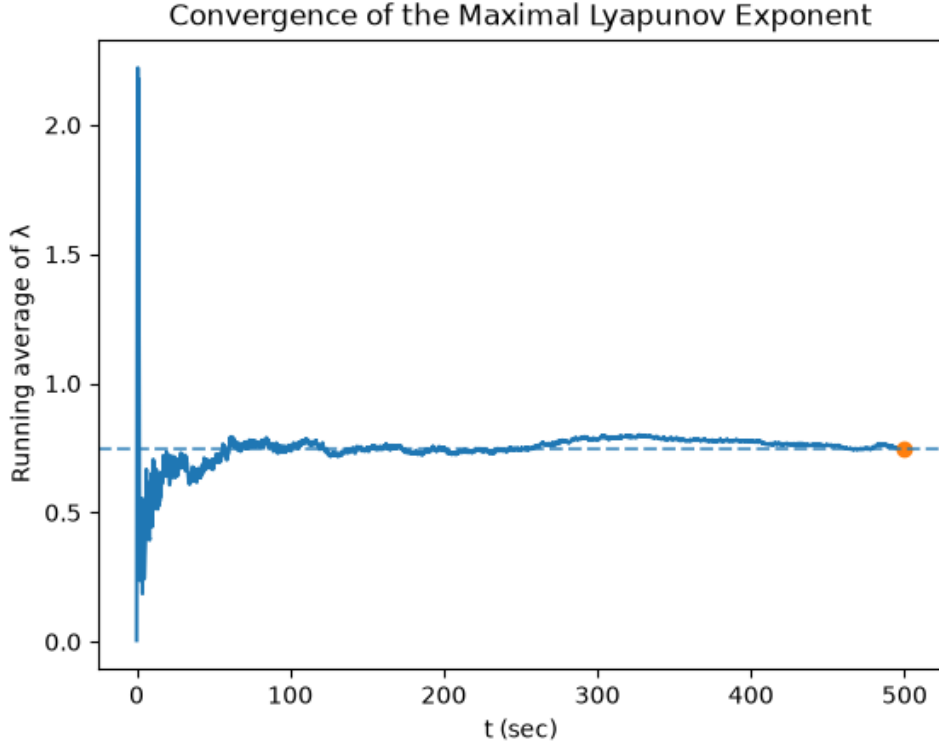
should approximately be a linear graph with slope  $\lambda$  where  $\lambda$  is the maximal lyapunov exponent of the system.

If  $|\mathbf{h}(t)|$  really does grow exponentially over time then for long term simulations, its magnitude would become arbitrarily large, causing the simulation to crash due to a floating point overflow error. To solve this issue and extract the maximal lyapunov exponent of the system, the benettin algorithm is used, which normalizes the perturbation vector at each step so its magnitude never becomes arbitrarily large while preserving its direction in state space. If  $d_k$  is the magnitude of the perturbation vector at the  $k$ th time step, then the maximal lyapunov exponent is given by:

$$\lambda = \frac{1}{T} \sum_k \ln(d_k)$$

where  $T$  is the time for which the simulation is run.

For the initial state  $\mathbf{r}(0) = (\frac{2\pi}{3}, 0, \frac{\pi}{4}, 0)$  with  $\mathbf{h}(0) = \frac{10^{-6}}{\sqrt{2}}(1, 0, 1, 0)$  and  $(m_1, l_1, m_2, l_2) = (5, 3, 7, 2)$ , the value of the maximal lyapunov exponent is found to be 0.7469. Since  $\lambda > 0$ , nearby trajectories in state space divergence exponentially.



### 3) Exploring global dynamics of the system

#### 3.1) State space reduction

The state space of the double pendulum is 4D. The total energy of the system is conserved and is given by:

$$E = \frac{m_1 l_1^2}{2} \omega_1^2 + \frac{m_2}{2} [l_1^2 \omega_1^2 + l_2^2 \omega_2^2 + 2l_1 l_2 \omega_1 \omega_2 \cos(\theta - \phi)] - m_1 g l_1 \cos \theta - m_2 g [l_1 \cos \theta + l_2 \cos \phi]$$

We can reduce the 4D state space to a 3D state space by using the total energy to find  $\omega_2$  as a function of  $\theta, \omega_1$  and  $\phi$

#### 3.2) Defining the goal

For creating Poincare sections, we need a 2D state space. Because the double pendulum has only one constant of motion, the total energy, we need to introduce another physically meaningful constraint to be able to create a 2D section of the 4D state space. One physically meaningful constraint would be to record  $\theta$  and  $\omega_1$  values only when  $\phi = 0$  and  $\omega_2 > 0$  i.e. study the motion of the main pendulum when the second pendulum crosses its lowermost point and has a positive angular velocity. To explore global behavior of the system under this constraint, we can integrate multiple trajectories at once, all of which satisfy  $\phi = 0$  and  $\omega_2 > 0$  at  $t = 0$ , and then store all the points  $(\theta, \omega_1)$  when  $\phi = 0$  and  $\omega_2 > 0$  for  $t > 0$  for each trajectory.

### 3.3) Finding a range for the initial values of $\theta, \omega_1$ such that $\phi = 0$ and $\omega_2 > 0$ is satisfied at $t = 0$

Imposing the constraint  $\phi = 0$  on the total energy at time  $t = 0$  we get:

$$E = \frac{m_1 l_1^2}{2} (\omega_1)_0^2 + \frac{m_2}{2} [l_1^2 (\omega_1)_0^2 + l_2^2 (\omega_2)_0^2 + 2l_1 l_2 (\omega_1)_0 (\omega_2)_0 \cos \theta_0] - m_1 g l_1 \cos \theta_0 - m_2 g [l_1 \cos \theta_0 + l_2]$$

$$E = \frac{(m_1 + m_2) l_1^2}{2} (\omega_1)_0^2 + \frac{m_2 l_2^2}{2} (\omega_2)_0^2 + m_2 l_1 l_2 (\omega_1)_0 (\omega_2)_0 \cos \theta_0 - (m_1 + m_2) g l_1 \cos \theta_0 - m_2 g l_2$$

Let  $A = \frac{m_2 l_2^2}{2}$ ,  $B = m_2 l_1 l_2 (\omega_1)_0 \cos \theta_0$ ,  $C = \frac{(m_1 + m_2) l_1^2}{2} (\omega_1)_0^2 - (m_1 + m_2) g l_1 \cos \theta_0 - m_2 g l_2$

So,

$$E = A(\omega_2)_0^2 + B(\omega_2)_0 + C$$

$$0 = A(\omega_2)_0^2 + B(\omega_2)_0 + C - E$$

Thus,

$$(\omega_2)_0 = \frac{-B \pm \sqrt{B^2 - 4A(C - E)}}{2A}$$

For  $(\omega_2)_0 \in \mathbb{R}$ ,

$$B^2 - 4A(C - E) \geq 0$$

So,

$$m_2^2 l_1^2 l_2^2 (\omega_1)_0^2 \cos^2 \theta_0 - 4 \left( \frac{m_2 l_2^2}{2} \right) \left( \frac{(m_1 + m_2) l_1^2}{2} (\omega_1)_0^2 - (m_1 + m_2) g l_1 \cos \theta_0 - m_2 g l_2 - E \right) \geq 0$$

Let  $-(m_1 + m_2) g l_1 \cos \theta_0 - m_2 g l_2 = V$  then,

$$m_2^2 l_1^2 l_2^2 (\omega_1)_0^2 \cos^2 \theta_0 - 4 \left( \frac{m_2 l_2^2}{2} \right) \left( \frac{(m_1 + m_2) l_1^2}{2} (\omega_1)_0^2 + V - E \right) \geq 0$$

Simplifying gives:

$$(\omega_1)_0^2 \leq \frac{2(E - V)}{l_1^2 (m_2 \sin^2 \theta_0 + m_1)}$$

Thus,

$$\begin{aligned} |(\omega_1)_0| &\leq \sqrt{\frac{2(E - V)}{l_1^2 (m_2 \sin^2 \theta_0 + m_1)}} \\ \Rightarrow -\sqrt{\frac{2(E - V)}{l_1^2 (m_2 \sin^2 \theta_0 + m_1)}} &\leq (\omega_1)_0 \leq \sqrt{\frac{2(E - V)}{l_1^2 (m_2 \sin^2 \theta_0 + m_1)}} \end{aligned} \quad (1)$$

Also since  $2A > 0$  always, for  $\omega_2 > 0$  we need  $-B + \sqrt{B^2 - 4A(C - E)} > 0$ .

Case 1) If  $B < 0$  then  $-B > 0$  and the inequality is satisfied automatically.

Case 2) If  $B \geq 0$  then  $-B \leq 0$ :

Thus,

$$\sqrt{B^2 - 4A(C - E)} > B$$

So,

$$B^2 - 4A(C - E) > B^2$$

(since  $B \geq 0$ )

Simplifying gives,  $C < E$

So,

$$\frac{(m_1 + m_2)l_1^2}{2}(\omega_1)_0^2 + V < E$$

$$(\omega_1)_0^2 < \frac{2(E - V)}{(m_1 + m_2)l_1^2}$$

Thus,

$$\begin{aligned} |(\omega_1)_0| &< \sqrt{\frac{2(E - V)}{l_1^2(m_1 + m_2)}} \\ \Rightarrow -\sqrt{\frac{2(E - V)}{l_1^2(m_1 + m_2)}} &< (\omega_1)_0 < \sqrt{\frac{2(E - V)}{l_1^2(m_1 + m_2)}} \end{aligned} \quad (2)$$

Combining the two inequalities (1) and (2) for the case  $B \geq 0$  we get:

$$\max \left( -\sqrt{\frac{2(E - V)}{l_1^2(m_1 + m_2)}}, -\sqrt{\frac{2(E - V)}{l_1^2(m_2 \sin^2 \theta_0 + m_1)}} \right) < (\omega_1)_0 < \min \left( \sqrt{\frac{2(E - V)}{l_1^2(m_2 \sin^2 \theta_0 + m_1)}}, \sqrt{\frac{2(E - V)}{l_1^2(m_1 + m_2)}} \right)$$

Which can be simplified to:

$$-\sqrt{\frac{2(E - V)}{l_1^2(m_1 + m_2)}} < (\omega_1)_0 < \sqrt{\frac{2(E - V)}{l_1^2(m_1 + m_2)}}$$

since  $m_2 \sin^2 \theta_0 + m_1 \leq m_1 + m_2 \forall \theta$ .

Also from both (1) and (2), for  $\omega_1$  to be a real number, we need  $E - V \geq 0$

So,  $E \geq V$

$$\Rightarrow \frac{-E - m_2 g l_2}{(m_1 + m_2) g l_1} \leq \cos \theta_0$$

Let

$$K = \frac{-E - m_2 g l_2}{(m_1 + m_2) g l_1}$$

Case 1) If  $K < -1$ , then  $\theta_0 \in [-\pi, \pi]$ .

Case 2) If  $-1 \leq K \leq 1$ , since  $\cos$  is symmetric,

$$-\cos^{-1}(K) \leq \theta_0 \leq \cos^{-1}(K)$$

Case 3) If  $K > 1$ , then no physically valid  $\theta_0$  exists.

So after choosing an energy value  $E$  if:

1) For  $K < -1$ :

$$-\pi \leq \theta_0 \leq \pi$$

For  $-1 \leq K \leq 1$ :

$$-\cos^{-1}(K) \leq \theta_0 \leq \cos^{-1}(K)$$

2) For  $B \geq 0$ :

$$-\sqrt{\frac{2(E-V)}{l_1^2(m_1+m_2)}} < (\omega_1)_0 < \sqrt{\frac{2(E-V)}{l_1^2(m_1+m_2)}}$$

For  $B < 0$ :

$$-\sqrt{\frac{2(E-V)}{l_1^2(m_2 \sin^2 \theta_0 + m_1)}} \leq (\omega_1)_0 \leq \sqrt{\frac{2(E-V)}{l_1^2(m_2 \sin^2 \theta_0 + m_1)}}$$

Then  $\phi = 0$  and

$$(\omega_2)_0 = \frac{-B \pm \sqrt{B^2 - 4A(C-E)}}{2A} > 0$$

is satisfied.

### 3.4) Creating and visually analyzing the Poincare sections

Numerically integrating  $\frac{d\mathbf{r}}{dt} = \mathbf{F}(\mathbf{r}(t))$  for 1000 trajectories, with  $(m_1, l_1, m_2, l_2) = (5, 3, 7, 2)$  and initial conditions as obtained in (3.3) and calculating the values of  $(\theta, \omega_1, \omega_2(\theta, \omega_1))$  when  $\phi = 0$  and  $\omega_2 > 0$  again for  $t > 0$  for each trajectory allows us to plot the Poincare section

$$\left( \theta, \frac{\partial L}{\partial \dot{\theta}} \right)$$

for various energy levels  $E$ . Giving each trajectory a color, we can see from the plots for  $E \in \{-450 \text{ J}, -245 \text{ J}, 0 \text{ J}, 245 \text{ J}, 490 \text{ J}\}$ , the presence of closed curves in the centre, with the curves becoming more and more defined as the energy increases. These closed curves can be identified with quasi-periodic orbits in phase space.

